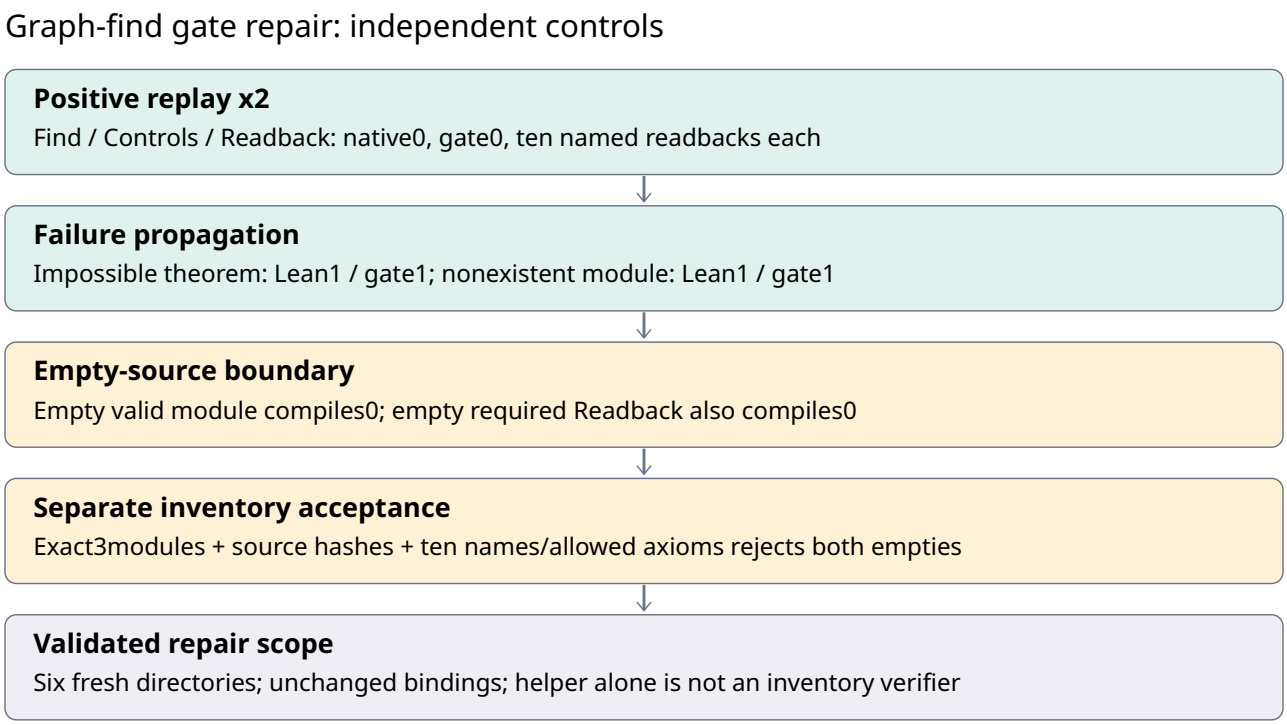
Independent review of graph-find gate repair



pencil-graph-gate-repair-review-20261101 | formal workflow controls | no scientific runs

The published repair propagates the tested native failures and creates distinct invocation snapshots. Two complete default invocations returned0 for Find, Controls and Readback, with ten expected axiom readbacks each. An impossible theorem and a nonexistent module both returned native1 and gate1. This validates the scoped failure-propagation and fresh-invocation repair. The helper itself does not verify a theorem inventory; that remains an explicit acceptance layer.

Bindings. Producer head e095d1e2a7120ef4c86f5b3117d92eb2c165c155 contains the graph-gate-repair packet and prospective gate-repair review joint. The helper, three mathematical modules and producer PLAN were extracted from that commit and matched their private producer originals. The helper SHA256 is9bd8c6370e7a50214036933c3b08372510c64d799829c868db0a2b6b90bafbba. All307 producer manifest entries matched. The existing producer snapshots were not overwritten. The published coordinator plan is copied as coordinator-review-plan.json; the local PLAN is an execution note written after the initial helper/report read, before controls.

Source inspection. Each call creates a new invocation directory, copies local Lean source and stable dependency source/binaries into a private snapshot, compiles the selected modules there, and writes per-module commands/stdout/stderr. Local compiled artifacts are not copied as substitutes for local source. Native failures stop the loop and set a nonzero gate status. Timeouts and caught exceptions have explicit status branches; terminal receipts use a temporary file followed by rename. Before/after hashes bind snapshot sources, non-output direct binaries, helper, PLAN and Lean executable. Early setup and final hashing errors occur outside the catch block and need not produce a terminal receipt; this review does not claim universal receipt production for every possible host failure. No forced timeout or concurrent-file-change test was performed.

Fresh positive checks. success1 and success2 each compiled exactly Find, Controls and Readback at native0/gate0. Their invocation directory names differ, all bindings stayed unchanged, and both named readback outputs include GraphFind.run_preserves, GraphFind.run_exists and the eight required FindControls declarations. The former depends only on propext; run_exists has no axioms; the controls use only propext and Quot.sound. These are fresh replays of unchanged mathematics, not new mathematical results.

Negative controls. ReviewerFalse contains example : False := by trivial; Lean rejected it and the repaired gate returned1 with a failure receipt. ReviewerDefinitelyMissingModule20261101 has no source or cached module; Lean returned1 and the gate returned1. ReviewerEmpty is a valid empty Lean source and correctly returned0. A separate stronger control replaced the required Readback.lean with an empty file while retaining the correct Find and Controls

files: all three compilations and the gate returned0, but there were no axiom readbacks. Every control used a fresh invocation directory; all six invocation receipts recorded unchanged bindings.

Explicit acceptance. The reviewer accept_inventory.py requires process/gate/native success, exact module order Find/Controls/Readback, unchanged bindings, byte-identical expected sources, and the exact ten distinct named readback lines with an allowed axiom set. It accepts both genuine default invocations and rejects the impossible theorem, missing module, arbitrary empty module and empty required Readback. The last case fails both source binding and declaration inventory. An empty file compiling successfully is normal Lean behavior; it must not be reported as a theorem verification success. This reviewer acceptance check is a private review artifact, not an alleged feature of gate.py or a deployed repository verifier.

Outcome and scope. Machine checks: passed for both default invocations and all four expected-boundary controls. Source fidelity: faithful_scoped for fresh invocation snapshots and tested native failure propagation. Review outcome: validated_scoped for that repair. Complete theorem acceptance is supplied separately here by source identity plus the named three-module/ten-readback check. The helper is not a complete named-declaration or axiom-policy verifier. Cached transitive Mathlib binaries and the compiler remain trusted dependencies; no independent binary kernel rechecker was used.

Independence. This reviewer identified the original wrapper defect and authored the older imported GraphAlgebra component, but did not author the new gate.py repair. The review is independent of this repair's producer, not fully blind to the prior problem or all imported mathematics. No scientific runs, cloud work, repository edits, or producer artifact changes occurred. The original defective helper and historical receipts remain preserved in their earlier packets.